Sports Analytics Final Report

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Introduction:

Tennis, one of the most popular sports in the world, allows players to combine athleticism, strategy, talent, skill, and mental strength in one-on-one competition (two-on-two for doubles). One of the main draws of the sport for players and spectators alike is the unpredictability of match outcomes. Like any sport, a large variety of factors can influence the outcome of a tennis match, but due to the one-on-one aspect of tennis, factors that may usually fly under the radar may have much larger influence. This project attempts to account for as many of these factors as possible, in order to create a machine learning model that can predict the outcome of professional tennis matches with adequate accuracy.

This project utilizes data from over 196,000 tennis matches from 1968-2026. Feature engineering was performed to better capture contexts such as ELO ratings, head-to-head, current form, performance trends, and more. Several machine learning methods, notably XGBoost, as well as a Neural Network, were deployed and tuned to develop a model that can adequately predict the outcome of professional tennis matches.

Related Work:

There have been various attempts to utilize machine learning methods for predicting match outcomes. In a 2020 study conducted by S. Wilkens et al., a wide range of machine learning techniques, such as logistic regression, random forests, and neural networks, were used to predict the outcome of tennis matches, yielding average results of 70% accuracy (S. Wilkens et al., 2020). In 2021, network analysis was deployed on a large dataset of tennis matches spanning about 30 years. “Evaluating the results shows the proposed methods provide more accurate predictions of tennis match outcome than classical approaches and outperform the existing methods in the literature and the current state-of-the-art models in tennis,” (Firas Bayram et al., 2021). A 2023 study aimed to determine if machine learning techniques can outperform standard ELO ratings when predicting tennis match results (Rory Bunker et al., 2023). Finally, an article published in 2015 states that, “We believe that the use of machine learning will lead to innovation in the field of tennis modelling,” (MEng Computing, 2015). While there is some thorough research on using machine learning for tennis use cases, there is still lots of room for exploration and improvement.

Methods:

Data:

The data was sourced from a GitHub Repository called TML-Database, which charts professional tennis matches and accompanying statistics. The repository provides data sets of matches for each year from 1968 to 2026. Important features included in each dataset include:

- Winner and loser names
- Date of the tournament
- Length of the match in minutes
- Surface (Clay, Hard, Grass)
- Round (QF, SF, F, etc.)
- Stats for winner/loser (aces, double faults, percentages, etc.)

Features for older matches are more sparse, particularly match statistics and player demographics. However, this is not a significant problem since these matches will not provide as much predictive power on future matches as more recent matches would, which has many more complete features.

Cleaning:

The initial step was to combine all datasets into one, which was done in Python, resulting in final dataset of almost 400,000 rows. The data was already relatively clean, but some extra steps were taken to ensure proper data types for dates and integers. It is also relevant to note that 2026 matches were tracked daily, while previous matches were tracked by the week of the tournament, so all of the data was standardized to the week of the tournament for consistency. Another necessary step that was taken was normalizing the names of players. Some players such as Alex de Minaur for example, were sometimes entered as Alex De Minaur, which would impact feature engineering.

A following important and necessary step was to double the dataset. Since the original dataset contained columns for winner, loser, and the outcome of the match, in order to avoid leakage, each match is recorded twice from the loser's perspective and winner's perspective. The players are now labeled as player and opponent, and after feature engineering, columns that indicate performance within the match itself were dropped, such as how long the match was and each player's statistics within the match.

Feature Engineering:

Feature engineering takes up the bulk of this project. By computing and engineering various new features, the dataset of about 50 columns was expanded to just over 215. New variables that were created include:

- Dummy variables for surface, round, tournament level, handedness, and indoors vs outdoors
- Each player's head-to-head record preceding the match, as well as win percentage against the opponent, and who won the most recent matchup
- ELO ratings for each player, which accounts for the strength of the opponent that each player defeats throughout the whole dataset
- ELO ratings for each surface, as well as a weighted and average ELO rating that accounts for overall and surface ELO
- Differences in ELO, age, height, rank, seed etc.
- Win/loss streaks
- Match statistics (aces, double faults, break points faced, serve percentage, etc.) across various match windows (1, 3, 5, 10, 30, 60, 90)
- Win rates across daily windows (30, 60, 90)
- Time on court within the current tournament
- Days of rest between tournaments
- Glicko ratings (ratings, rating deviation, volatility)

These features provided an adequate starting point for collecting nuance and context for each match. It is necessary to note that ensuring proper sorting of match date and round at each iteration of feature calculation was imperative to avoiding leakage from test to training data.

Model Training and Tuning:

Seven machine learning models were trained to predict the outcome of professional tennis matches. The models trained include:

- Baseline Logistic Regression
- Baseline Decision Trees
- Baseline Random Forests
- Baseline XGBoost
- Tuned XGBoost trained on all data
- Tuned XGBoost trained on matches starting in 2011
- Neural Network

All of the models, with the exception of the 2011 trained XGBoost, were trained on all of the available data, starting from 1968 through 2024. The second tuned XGBoost model was trained on all matches from 2011 through 2024, to see if more match data provided model improvements, or simply contributed noise, and did not generalize to today's game of tennis.

Both XGBoost models were tuned using random search. The fully trained model implemented a random search with 50 iterations and five time series cross validation splits, while the shortened model's random search was conducted with 200 iterations and 10 time series cross validation splits. The parameters selected for each model are as shown:

	Full XGBoost	Shortened XGBoost
Subsample	1.0	1.0
N_estimators	500	300
Min_child_wieght	7	5
Max_depth	6	12
Learning_rate	0.05	0.1
Gamma	0	0.3
Colsample_bytree	0.5	0.1

Both sets of parameters were used to re-train the two XGBoost models, one trained on all data and one trained on 2011-present data, and it was determined that the parameters for the shortened XGBoost model yielded better performance in both models. However, this most likely stems from the lengthened iterations and time series splits, as opposed to the data split. After optimal parameters were established, and adequate performance on the test set was achieved, the final model was trained on all available data, from 1968 through early 2026, to hopefully predict upcoming matches with reasonable accuracy.

The neural network was trained using the Keras API from TensorFlow. The Neural Network was tuned with a random search to find optimal values for hyperparameters such as number of layers and filters, dropout rate, and batch size. The random search was run for 200 iterations, each iteration training for 50 epochs, and an initial learning rate of 0.001. Early stopping was implemented to ensure that training was halted if AUC did not improve for five epochs. Learning rate reduction was also implemented, which reduced the learning rate by half if an improvement on AUC was not made after three epochs. Each iteration trained a Neural Network on all match data prior to 2022, and validated performance on all matches in 2022. The full model summary of the tuned Neural Network's architecture is featured in the appendix.

Evaluation:

As previously stated, with exception of the shortened tuned XGBoost model, each model was trained on all matches from 1968 through 2024, and tested on matches from 2024 up to right before the 2026 Australian Open. Each model's performance was evaluated based on its accuracy and AUC scores on the testing set, results of which will be displayed in the discussion section.

Prediction:

The purpose of this project is to develop a machine learning model that can predict the outcome of professional tennis matches with reasonable accuracy. The model was used to predict the match outcomes of the three largest tournaments that succeed the training data; 2026 Australian Open, Dallas Open, and Rotterdam Open.

<u>Discussion:</u>

Various Models:

The performance of each model on the testing set is displayed below:

	Model	Accuracy	AUC
0	Logistic Regression	65.07%	72.91%
1	Decision Trees	57.53%	57.53%
2	Random Forests	65.18%	71.14%
3	XGBoost	64.36%	70.60%
4	Tuned XGBoost 1968	66.05%	72.64%
5	Tuned XGBoost 2011	66.15%	72.54%
6	Neural Network	66.43%	73.02%

The baseline Decision Trees model is the obvious worst performer, with every other model achieving an AUC of at least 70%. It is interesting to note that both the Logistic Regression and Random Forests models performed better than the baseline XGBoost model. However, both tuned XGBoost models, and the Neural Network are the clear best performers, all achieving similar results with accuracies above 66%, and AUCs above 72.5% on the testing data. The Neural Network outperforms all other models in both accuracy and AUC, followed closely by the tuned XGBoost models which are almost identical.

Important Features:

Generating SHAP graphs for both the full XGBoost model and Neural Network led to some interesting insights on what the models value most for prediction. The XGBoost model trained on all data heavily relies on ELO ratings, specifically difference in ELO and surface ELOs, for its predictions. The XGBoost model also places importance on the number of matches played in the last 30 days by the player, with lower values tending to lead the model to predict a loss. This makes sense as more matches played indicates that the player is winning most matches, and is likely in good form.

The Neural Network however, places most of its predictive power on Glicko ratings and deviations. While the Neural Network still values ELO ratings, it is interesting that it finds Glicko rating, as well as deviation of that rating, the better predictor of match results. It is also interesting to point out that players who are right-handed tend to have a higher chance of winning the match, however this is most likely due to the fact that the majority of tennis players are right handed.

See the appendix for SHAP graphs.

Prediction Results:

Using the Neural Network to predict upcoming tournament results of three tournaments, the results are as follows:

	Tournament	Accuracy	AUC
0	Australian Open 2026	77.95%	84.06%
1	Dallas Open 2026	75.81%	80.85%
2	Rotterdam 2026	75.81%	84.91%

The model performed relatively well across all three tournaments, with accuracies greater than 75% and AUCs greater than 80%. Better performance for the Australian Open can most likely be explained by the larger sample size of matches. It is important to note that the model evaluated each match on an individual level, not considering the results of previous matches within that tournament.

It is also relevant to mention that the model correctly predicted Carlos Alcaraz to win all of his matches in the Australian Open, but fell short with the Dallas Open and Rotterdam, with the actual winners, Ben Shelton and Alex de Minaur respectively, predicted to lose in the final.

While these are still promising results, several caveats remain. The model predicted each match based on data from before the tournament. It does not take into account results from previous rounds within the tournament, which are extremely important when predicting matches in later rounds. A true tournament simulation, with features recalculated after each round, would ideally yield the model's true potential in predicting tennis matches and tournaments. The model also heavily relies on the player's ratings (ELO and Glicko) when making its predictions, indicating that it will usually select the higher rated player to win and still achieve reasonable accuracy. Continuing to add features that represent deeper context could potentially help the model correctly predict more upsets, and add several percentage points in accuracy.

Upset Pitfalls:

The major pitfall of the model is that it has difficulty with upsets, especially with new players it does not have much data on. Below is a perfect example, which shows the model's confidence in its predictions that it got incorrect. Focus on the second matchup between Sebastian Korda and Michael Zheng:

	player	opponent	predicted_prob
394314	jiri lehecka	arthur gea	0.942533
394226	sebastian korda	michael zheng	0.876767
394443	jannik sinner	novak djokovic	0.808748

The model was quite confident that Sebastian Korda would beat Michael Zheng, and from the data given to the model, it cannot be faulted too much for thinking that. However, what the model cannot account for is the fact that Sebastian Korda, who has now been on the tour for several years, has not met expectations, and has had some very disappointing results. Michael Zheng on the other hand, was the number one ranked NCAA tennis player at the time of the match and is fresh off back-to-back NCAA national championship titles. This is his first Grand Slam match, so expectations are low for him. Michael Zheng has very little pressure and is in form, while Sebastian Korda has quite a bit of pressure, and is in questionable form. With this context in mind, it is not difficult to see why Michael Zheng could have defeated Sebastian Korda. However, the model does not have past data for Michael Zheng, and can only see that Sebastian Korda is the higher ranked player and overwhelming favorite. Context like this is very important in tennis, and hard to implement in machine learning models.

The model also cannot be faulted for predicting Jannik Sinner to beat Novak Djokovic. Nobody saw that upset coming.

Conclusion and Future Work:

In conclusion, this project trained and implemented multiple models to predict outcomes of professional tennis matches. Out of the seven models trained, the tuned XGBoost and Neural Network models performed the best on the validation data, all achieving test accuracy of above 66% and test AUC above 72.5%. When using the Neural Network model trained on all data to predict the outcome of three tournaments in 2026, accuracy and AUC averaged around 76% and 82% respectively. It is interesting to point out the features that the models focus on, with player ratings (ELO and Glicko) being the main predictors.

While continuing to tune the models further could lead to marginal improvement, it is most likely that most of the improvement will be made by creating more and better features. While the features currently included in the model do a good job of capturing player performance, it does not do as well accounting for context and player matchups in regard to play styles. Examples of features that could account for play styles are:

- Maximum and average speed of shots (serve, forehand, backhand etc.)
- Surface head-to-head
- Serve tendencies (slice, kick, tee, wide)
- Return tendencies
- Winner and unforced errors
- Speed of player
- Win rates within rally windows (< 5 shots, 5 – 10 shots, > 10 shots, etc.)
- Shot selection (dropshots, slice,
- Court position
- Top spin rates
- Other player ratings such as TruSkill, and Colley

Implementing new features such as these that represent a player’s play style could potentially significantly improve its performance and ability to predict match outcomes, as it could now analyze matches from the perspective of player vs player on a more granular level.

Accounting for injuries would also make a huge impact on the model's performance. Information such as injury history and susceptibility is vastly important. A great example is in tournaments played in hot weather, where players prone to cramping are at a disadvantage. Injuries, however, are hard to predict and can happen at any time, so much more data and features would be needed to begin accurately capturing their effects and risks.

Finally, conducting a tournament simulation, for example the 2026 Australian Open, would allow us to see how well the model could perform when predicting matches round by round. In practice, this is difficult, as it would require the recalculation of features after each round, which requires a large chunk of past data. While time and computationally expensive, this is possible to achieve with adequate time.

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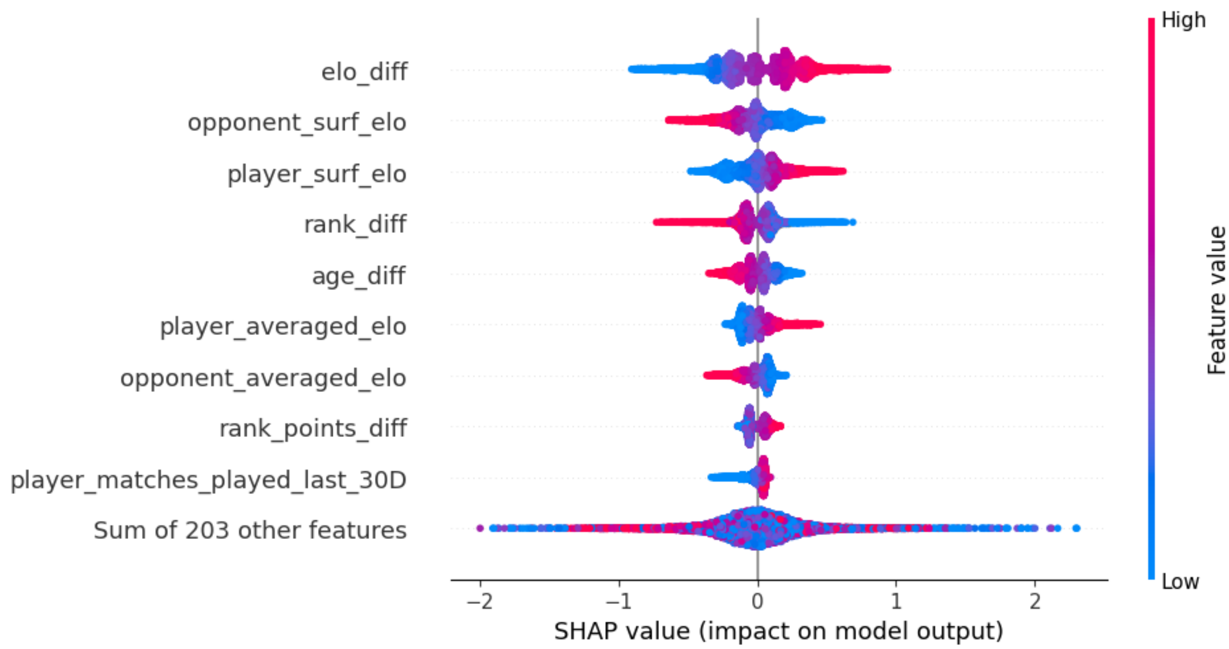
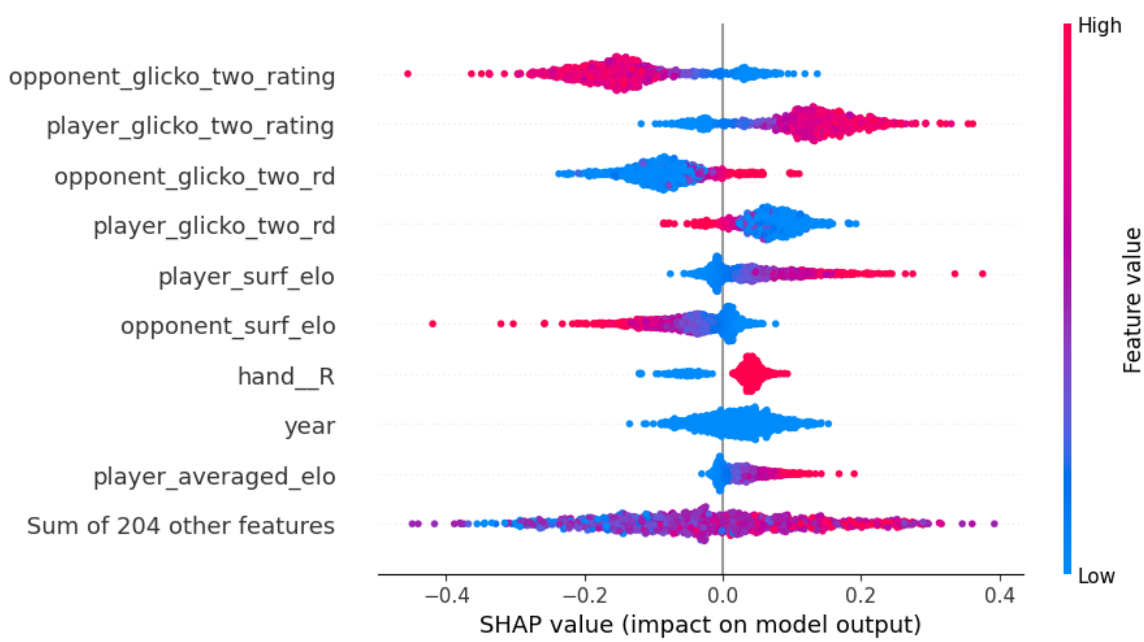
Appendix:

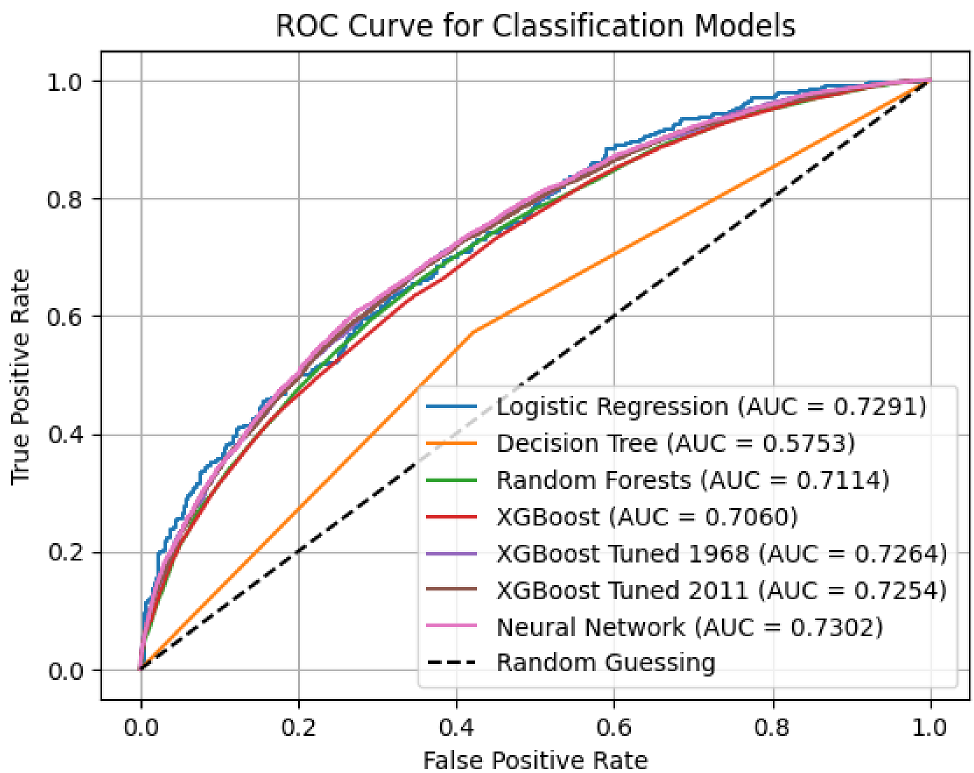
XGBoost Model Trained on All Data SHAP Graphs:



Neural Network SHAP Graphs:



ROC Comparisons of All Models:



Neural Network Model Summary:

Layer (type)	Output Shape	Param #
dense (Dense)	(None, 256)	54,784
batch_normalization (BatchNormalization)	(None, 256)	1,024
dropout (Dropout)	(None, 256)	0
dense_1 (Dense)	(None, 128)	32,896
batch_normalization_1 (BatchNormalization)	(None, 128)	512
dropout_1 (Dropout)	(None, 128)	0
dense_2 (Dense)	(None, 64)	8,256
batch_normalization_2 (BatchNormalization)	(None, 64)	256
dropout_2 (Dropout)	(None, 64)	0
dense_3 (Dense)	(None, 32)	2,080
batch_normalization_3 (BatchNormalization)	(None, 32)	128
dropout_3 (Dropout)	(None, 32)	0
dense_4 (Dense)	(None, 32)	1,056
batch_normalization_4 (BatchNormalization)	(None, 32)	128
dropout_4 (Dropout)	(None, 32)	0
dense_5 (Dense)	(None, 32)	1,056
batch_normalization_5 (BatchNormalization)	(None, 32)	128
dropout_5 (Dropout)	(None, 32)	0
dense_6 (Dense)	(None, 1)	33